

PROJECT PROPOSAL

Machine Learning-Enhanced UWB Ranging Calibration

DWM1000 Time-of-Flight Sensor Calibration via Supervised Regression
An Industry 4.0 Approach

Graduate Program - Industrial & Systems Engineering
ISE 572 - Industry 4.0 Machine Learning
Spring 2026

Hardware: 2x DWM1000 + Arduino Nano 33 BLE Sense Lite
Protocol: SS-TWR, Channel 5 (6489.6 MHz), PRF 64 MHz
Standard: IEEE 802.15.4-2011 UWB PHY
Target: 10 cm MAE indoors via ML post-correction

Section 1: Problem Statement

Ultra-Wideband (UWB) real-time location systems (RTLS) are a cornerstone technology of Industry 4.0, deployed across manufacturing, warehouse logistics, and healthcare to enable centimeter-level indoor positioning where GPS cannot operate. Companies including BMW, Volkswagen, and Amazon use UWB RTLS to track automated guided vehicles (AGVs) on factory floors, enforce geofenced safety zones around heavy machinery, and locate high-value assets in warehouses. The global UWB market is projected to exceed \$4.8 billion by 2030 (Allied Market Research), driven largely by Industry 4.0 adoption. The Decawave DWM1000, an IEEE 802.15.4-2011 compliant UWB transceiver manufactured by Qorvo, is one of the most widely deployed chipsets in industrial RTLS, with a manufacturer-specified precision of 10 cm indoors.

The specific Industry 4.0 problem this project addresses is the **persistent gap between specified and real-world UWB ranging accuracy** caused by systematic biases that the standard single-point antenna delay calibration cannot correct. The DWM1000 measures distance via Single-Sided Two-Way Ranging (SS-TWR): an initiator sends a poll message, the responder replies, and the time-of-flight (ToF) is computed from the round-trip time minus the known reply delay. While the antenna delay register compensates for internal propagation delays, five distinct error sources remain uncorrected:

- **Distance-dependent bias:** Near-field antenna coupling and clock skew produce different error characteristics at 1 m versus 10 m. Our preliminary data demonstrates this directly: after applying the manufacturer's recommended antenna delay of 16700 (for Channel 5 / PRF 64 MHz), the system still reports a -18.7 cm mean bias at 5.01 m with a standard deviation of 35.5 cm, far exceeding the 10 cm specification.
- **Signal power coupling:** The DW1000's leading-edge detection algorithm behaves differently at different received power levels. Strong signals produce crisp first-path timestamps; weak signals at range cause the noise floor to rise, systematically degrading timestamp quality. Static calibration ignores this power-dependent correction entirely.
- **Multipath and NLOS conditions:** In cluttered industrial environments, reflected signals arrive later than the direct path, always positively biasing the ToF estimate. The DW1000 User Manual (Section 4.7) documents that when the first-path to received power ratio exceeds 10 dB, NLOS conditions are likely and errors can exceed 1 meter.
- **Temperature drift:** The DW1000 datasheet specifies approximately 2.15 mm/C of ranging drift due to crystal oscillator frequency shift. Over the 10 to 15 C ambient swings typical of warehouses and production floors between morning startup and peak operation, this accumulates to 2 to 3 cm of uncompensated error.
- **Antenna orientation:** The DWM1000's integrated chip antenna has a non-isotropic radiation pattern. Rotating the module changes the effective path delay by several centimeters, causing positioning errors when tracked assets like forklifts or carts change heading.

The cost of not solving this problem is both quantifiable and consequential. A single AGV collision caused by positioning error can cost \$50,000 to \$200,000 in equipment damage, production downtime, and safety investigation. Safety zone violations around industrial robots

go undetected when positioning uncertainty exceeds the geofence margin. Current industry practice requires specialized RF engineers for manual site-specific calibration, typically 2 to 3 days at \$150 to \$250/hour per facility. For a company operating 20 to 50 facilities, annual recalibration costs alone can exceed \$300,000.

This project applies supervised machine learning to learn a correction function that maps raw DWM1000 sensor readings (distance, signal power, channel diagnostics, temperature) to true distances, addressing all five error sources simultaneously. The result is a portable, automated calibration pipeline directly applicable to manufacturing, logistics, and healthcare RTLS deployments, reducing calibration from days of expert labor to hours of structured data collection followed by automated model training.

Section 2: Data Considerations

This project uses **real sensor data collected from physical DWM1000 UWB hardware**, not simulated or pre-existing datasets. We have built and validated a complete data collection pipeline using two Arduino Nano 33 BLE Sense Lite boards (nRF52840 SoC), each wired to a DWM1000 transceiver module via SPI. The system performs Single-Sided Two-Way Ranging at Channel 5 (6489.6 MHz center frequency, 499.2 MHz bandwidth), and the initiator outputs 11 columns of CSV-formatted data over USB serial at 115200 baud at a 20 Hz sampling rate.

Per-sample feature vector (11 raw columns plus derived features):

Feature	Description	Source
distance_m	Raw SS-TWR distance (meters, float32)	DW1000 ToF calculation
rx_power_dBm	Total received signal power (dBm, float32)	DW1000 register 0x12
fp_power_dBm	First-path signal power (dBm, float32)	DW1000 register 0x15
quality	Preamble accumulation quality index (float32)	DW1000 RX_FQUAL
tof_ticks	Raw time-of-flight (40-bit DW1000 clock ticks)	DW1000 timestamps
temp_C	Ambient temperature (Celsius, float32)	LPS22HB on-board sensor
pressure_hPa	Barometric pressure (hPa, float32)	LPS22HB on-board sensor

Additional diagnostic registers through the BLE pipeline include `std_noise`, `fp_ampl1/2/3`, `cir_power`, and `rxpacc`. From these, we engineer three derived features: **power_diff_dB** (`rx_power` minus `fp_power`, the primary NLOS indicator per DW1000 User Manual Section 4.7), **channel_condition** (a 3-class label encoding LOS/Marginal/NLOS based on the manufacturer's 6 dB and 10 dB thresholds), and **confidence** (a 0 to 1 quality score combining preamble quality and channel state). Each sample is labeled with the ground-truth distance measured by laser distance meter and the antenna orientation angle.

Data collection plan and volume: A structured campaign spanning 7 reference distances (0.5, 1.0, 2.0, 3.0, 5.0, 7.0, and 10.0 m) with 4 antenna orientations (0, 90, 180, 270 degrees) per distance. Each trial collects 500 measurements at 20 Hz, yielding approximately **14,000 labeled samples across 28 distinct conditions**. Three preliminary runs have been completed (332

samples at 1 m, 500 at 2 m, 100 at 5.01 m), validating the pipeline end to end. In a production deployment, the same data comes from installed UWB infrastructure, with anchors at known positions providing continuous ground truth.

Anticipated data challenges and mitigations:

- **Temporal correlation and data leakage:** Samples within a single 25-second trial are autocorrelated and share identical true distance and environmental conditions. Random train/test splitting would leak this structure, producing falsely optimistic metrics. *Mitigation:* All validation uses trial-level splitting, and our primary evaluation metric is Leave-One-Distance-Out (LODO) cross-validation, which holds out all samples at an entire reference distance per fold.
- **Class imbalance in channel conditions:** Preliminary testing in a typical indoor environment showed 99% of samples classified as NLOS (fp_rx_ratio below -10 dB), with only 1% LOS or marginal. The classifier cannot learn LOS correction if it rarely sees LOS data. *Mitigation:* We will collect dedicated LOS trials in open environments (outdoor courtyard, large open room with no nearby reflective surfaces within 2 m) to ensure balanced representation across channel conditions.
- **Environmental variability and domain shift:** Temperature, humidity, and multipath environment vary across locations and time of day. A model trained in one room may degrade in another. *Mitigation:* Temperature and pressure are included as input features so the model can learn environmental compensation directly. LODO cross-validation tests interpolation to unseen distances, and we plan to validate in at least two distinct indoor environments.
- **Labeling accuracy:** Ground truth distance is measured manually with a tape measure or laser meter, introducing potential labeling error of 1 to 2 cm. *Mitigation:* Using a laser distance meter (rated to 1.5 mm accuracy) for all final data collection, and averaging 500 samples per trial to reduce the impact of any single measurement outlier.

Section 3: Proposed ML Solution

This is fundamentally a **supervised regression problem**: given a feature vector of raw sensor readings from a single UWB measurement, predict the continuous-valued true distance between the two modules. We chose regression over classification because the output (distance in meters) is inherently continuous, and discretizing it into bins would discard precision. We frame the target as predicting the true distance directly rather than predicting the signed error term (true minus raw), because direct prediction proved more numerically stable in initial experiments and avoids compounding errors when the raw measurement contains large outliers.

Alternative approaches considered: A Kalman filter could smooth temporal noise but cannot learn distance-dependent or power-dependent systematic biases. Deep learning (ANNs/CNNs) was considered, but with only 8 to 12 structured features and 14,000 samples, tree-based ensemble methods are known to outperform neural networks on tabular regression. Pure physics-based correction (applying the 2.15 mm/C temperature coefficient) addresses only one of five error sources. The multi-stage ensemble approach below combines classification and regression to handle all error sources simultaneously.

Model output specification: The primary model outputs a single float32 value representing the corrected distance in meters. For a raw reading of 4.823 m at 5.01 m true distance (our actual baseline data), the model should output approximately 5.01 m, correcting the -18.7 cm bias. The output is bounded: predictions below 0.0 m are clipped to 0.0, and predictions exceeding the raw reading by more than 50% are flagged as unreliable (edge case handling for extreme multipath). The secondary NLOS classifier outputs one of three discrete classes: LOS (0), Marginal (1), or NLOS (2), following the DW1000 User Manual's documented thresholds.

Our solution implements a **multi-stage ML pipeline** with three distinct stages:

Stage 1 - Channel Condition Classification (Random Forest Classifier): Before distance correction, a Random Forest classifier categorizes each measurement as LOS, Marginal, or NLOS using signal diagnostics (rx_power, fp_power, quality, amplitude ratios). The DW1000 User Manual (Section 4.7) provides the physical basis: first-path to total power difference below 6 dB indicates LOS, above 10 dB indicates NLOS. This classification is fed as an input feature to the regressor, telling it which correction regime to apply.

Stage 2 - Distance Correction (Regression, three models compared): The core stage predicts true distance from raw measurements plus channel condition and confidence. We train and compare three regression approaches covered in the course:

- **Linear Regression with Ridge Regularization:** The interpretable baseline, establishing whether correction is predominantly linear with distance and power. Polynomial features (degree 2 to 3) capture the known quadratic power-distance relationship. Ridge regularization prevents overfitting on interaction terms. Fast inference (sub-microsecond) and fully interpretable coefficients.
- **Random Forest Regressor:** Captures non-linear interactions (e.g., signal power behaves differently at 1 m versus 10 m). Inherently robust to multipath outliers because tree splits are rank-based. Provides feature importance rankings that validate whether physics-motivated features carry predictive signal.
- **Gradient Boosted Trees (GBR):** Expected best performer for structured tabular regression. Iteratively fits trees to residual errors, naturally handling heteroscedastic noise where variance increases with distance. Key hyperparameters (learning_rate=0.05, n_estimators=300, max_depth=4, subsample=0.8) tuned via LODO cross-validation.

Stage 3 - Confidence Scoring: Each corrected measurement receives a confidence weight (0 to 1) derived from signal quality and channel condition. In multi-anchor trilateration, this enables weighting high-confidence measurements more heavily, improving the overall position solution.

Cross-validation strategy: To prevent data leakage from temporal correlation within trials, we use Leave-One-Distance-Out (LODO) cross-validation: each fold holds out all samples at one reference distance and trains on the remaining six. This directly tests interpolation to unseen distances. All splits are at the trial level, never individual samples.

Section 4: Expected Impact

The primary impact is transforming UWB calibration from a manual, expert-dependent process into an automated, data-driven pipeline deployable by non-specialists at any facility. This delivers measurable benefits across three dimensions: **accuracy, cost, and safety.**

Accuracy improvement: Our baseline data shows -18.7 cm mean bias and 35.5 cm standard deviation at 5.01 m with static calibration alone. The ML pipeline targets a mean absolute error (MAE) of 10 cm or less across 0.5 to 10 m, representing at least a 50% reduction in positioning error. For AGV lane-following with 30 cm lane margins, this moves the system from frequent out-of-lane events to reliable navigation.

Deployment cost reduction: Replacing 2 to 3 days of on-site RF engineering (\$2,400 to \$6,000 per facility at \$150 to \$250/hour) with 2 hours of structured data collection plus automated training could save an estimated \$2,000 to \$5,500 per facility deployment. For organizations operating dozens of sites, this translates to annual savings exceeding \$100,000 in calibration labor alone.

Safety improvement: NLOS detection flags degraded measurements instead of silently reporting inaccurate positions. In a safety-critical application where a 1 m positioning error could place a worker inside a robot's operating envelope, the confidence scoring system provides the downstream safety logic with measurement quality information that static calibration cannot.

Adaptive operation: Models incorporating temperature and signal power features compensate for environmental variation in real time, maintaining accuracy through 10+ C daily temperature swings without manual recalibration.

Success metrics (ML performance and operational):

Metric	Type	Target	Why It Matters
MAE	ML Performance	10 cm or below	Primary accuracy measure
RMSE	ML Performance	15 cm or below	Penalizes large outliers
LODO MAE	ML Generalization	15 cm or below	Tests unseen distances
NLOS Accuracy	ML Classification	85%+	Channel condition detection
Inference Rate	Operational	20 Hz or above	Matches sensor output rate
Calibration Time	Business/Operational	Under 3 hours total	vs. 2 to 3 days manual
Cost per Site	Business/Financial	Under \$500 labor	vs. \$2,400 to \$6,000 manual

The project deliverables include a complete Python codebase hosted on GitHub, trained models for real-time deployment, and a technical report with reproducible results. The pipeline is extensible to other UWB chipsets and multi-anchor trilateration for 2D/3D positioning.